

Assessment Test: Software Tester (QA)

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1. Test Case Design

A. User Registration

ID	Test Scenario / Test Case	Type	Expected Result
REG-01	Register with valid name, email and password	Positive	Account should be created successfully.
REG-02	Submit registration with all mandatory fields empty	Negative	Required-field validation should be displayed.
REG-03	Enter invalid email format	Negative	Invalid email error should be displayed.
REG-04	Register using an already registered email	Negative	Duplicate-email error should be displayed.
REG-05	Enter password below minimum allowed length	Edge/Negative	Password validation should be displayed.
REG-06	Enter password meeting all requirements	Positive	Registration should succeed.
REG-07	Password and Confirm Password do not match	Negative	Password mismatch validation should appear.
REG-08	Enter very long name/email/password	Edge	Application should handle input safely.
REG-09	Enter special characters/script in input fields	Negative/Security	Input should be safely handled/escaped.
REG-10	Submit registration multiple times quickly	Edge	Duplicate account should not be created.

B. Login

ID	Test Scenario / Test Case	Type	Expected Result
LOGIN-01	Login with valid email and password	Positive	User should successfully log in.
LOGIN-02	Login with valid email and wrong password	Negative	Login should fail with proper error.
LOGIN-03	Login with unregistered email	Negative	Appropriate error should be displayed.
LOGIN-04	Submit with email and password empty	Negative	Required-field validation should appear.
LOGIN-05	Enter invalid email format	Negative	Invalid email message should appear.
LOGIN-06	Check password visibility	Positive	Password should be masked.
LOGIN-07	Enter very long email/password	Edge	Application should handle input correctly.
LOGIN-08	Enter spaces before/after email	Edge	Application should handle/trims spaces appropriately.
LOGIN-09	Repeated wrong-password attempts	Negative/Security	Excessive failed attempts should be controlled.
LOGIN-10	Access task page without login	Negative/Security	User should be redirected/denied access.

C. Create Task

ID	Test Scenario / Test Case	Type	Expected Result
TASK-C-01	Create task with valid data	Positive	Task should be created and displayed.
TASK-C-02	Create task with empty title	Negative	Required-field validation should appear.
TASK-C-03	Create task with empty optional description	Edge	Task should save if description is optional.
TASK-C-04	Enter maximum allowed characters	Edge	Task should save successfully.
TASK-C-05	Enter extremely long task data	Edge	Application should handle input without crashing.
TASK-C-06	Enter special characters in task	Positive/Edge	Valid special characters should display correctly.
TASK-C-07	Click Save multiple times	Edge	Duplicate tasks should not be created.

D. View Task List

ID	Test Scenario / Test Case	Type	Expected Result
TASK-V-01	Open task list after login	Positive	User's tasks should be displayed.
TASK-V-02	User with no tasks opens list	Edge	Appropriate empty-state message should appear.
TASK-V-03	Create multiple tasks and view list	Positive	All created tasks should appear.
TASK-V-04	Refresh task list	Positive	Previously saved tasks should remain available.
TASK-V-05	Try to view another user's tasks	Negative/Security	Other user's private tasks should not be visible.

E. Edit Task

ID	Test Scenario / Test Case	Type	Expected Result
TASK-E-01	Edit task with valid information	Positive	Updated information should be saved.
TASK-E-02	Edit task with required field empty	Negative	Validation error should appear.
TASK-E-03	Edit task with maximum allowed input	Edge	Task should update successfully.
TASK-E-04	Cancel editing	Positive	Original task data should remain unchanged.
TASK-E-05	Refresh after editing	Positive	Updated data should persist.

F. Delete Task

ID	Test Scenario / Test Case	Type	Expected Result
TASK-D-01	Delete an existing task	Positive	Task should be removed from the list.
TASK-D-02	Delete task and refresh page	Positive	Deleted task should not reappear.
TASK-D-03	Cancel delete confirmation	Positive	Task should remain.
TASK-D-04	Try deleting another user's task	Negative/Security	Unauthorized deletion should be prevented.
TASK-D-05	Double-click Delete	Edge	No unexpected duplicate request/error should occur.

G. Input Validation & Error Handling

ID	Test Scenario / Test Case	Type	Expected Result
VAL-01	Submit forms with blank mandatory fields	Negative	Clear validation messages should appear.
VAL-02	Enter invalid email	Negative	Email validation should be shown.

ID	Test Scenario / Test Case	Type	Expected Result
VAL-03	Enter extremely long input	Edge	Input should be restricted/handled safely.
VAL-04	Enter HTML/script input	Security	Application should safely handle the input.
VAL-05	Database/server unavailable during task creation	Negative	Friendly error should be displayed.
VAL-06	Network failure during save/update/delete	Negative	Operation should fail gracefully.
VAL-07	Refresh after successful operation	Positive	Correct saved state should remain.
VAL-08	Invalid/expired session	Negative	User should be redirected to login.
VAL-09	Unexpected API error	Negative	Technical details should not be unnecessarily exposed.
VAL-10	Multiple rapid submissions	Edge	Duplicate requests should be prevented/handled.

2. Potential Bugs / Risk Areas

#	Potential Bug / Risk	Severity	Reason / Impact
1	User can access task list without authentication	Critical	Unauthorized users may access protected data.
2	One user can view another user's tasks	Critical	Serious privacy/data leakage issue.
3	One user can edit/delete another user's task	Critical	Unauthorized modification or deletion of data.
4	Password stored or transmitted insecurely	Critical	User credentials could be compromised.
5	Duplicate tasks created by multiple Save clicks	Major	Incorrect/duplicate database records may be created.
6	Deleted task reappears after refresh	Major	Indicates database/state synchronization problem.
7	Invalid input accepted without validation	Major	Can result in bad data and potential security problems.
8	Application crashes when database/server is unavailable	Major	Poor reliability and bad user experience.
9	Long/special-character input breaks task list UI	Minor	UI/data-display problems affect usability.
10	Login gives unclear/wrong error messages	Minor	Makes troubleshooting difficult and reduces usability.

3. Testing Approach

Priority 1 – Critical: Registration, login/authentication, authorization, user data isolation and task CRUD security.

Priority 2 – Major: Input validation, database persistence, error handling, duplicate submissions and session management.

Priority 3 – Minor: UI validation messages, boundary values, formatting and usability issues.

4. Final QA Conclusion

Testing should focus not only on whether the application works, but also on whether user data is secure and whether operations remain reliable under invalid, unexpected and boundary conditions.

Release recommendation: Before production release, verify authentication/authorization, user data isolation, CRUD persistence, validation, error handling, regression after fixes and basic security controls. Critical security and data-integrity issues must be resolved before release.